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Ultrafast carrier dynamics in a monolayer MoS_2 at carrier densities well above Mott density

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Abstract

Due to the growing interest in monolayer (ML) molybdenum disulfide (MoS_2) in several optoelectronic applications like lasers, detectors, sensors, it is important to understand the ultrafast behavior of the excited carriers in this material. In this article, a comprehensive study of the charge carrier dynamics of a monolayer MoS_2 flake has been studied using transient transmission technique near A-exciton under high excitation densities well above the Mott density. Fluence dependent studies has been carried out to understand the origin of the processes which modifies its optical response under excitation. The dissociation of excitons leads to an observed fast bandgap renormalization. At later times when large number of carriers relax the remaining carriers forms excitons leading to a bleaching effect.

Keywords: monolayer MoS_2, band gap renormalization, ultrafast carrier dynamics

S Supplementary material for this article is available online

(Some figures may appear in colour only in the online journal)

1. Introduction

In recent years, transition metal dichalcogenides (TMDC) have emerged as one of the important class of material for optoelectronic applications [1–5]. Many monolayer (ML) TMDC materials have a direct bandgap nature which can be easily altered to an indirect nature just by adding another layer on top of it [6, 7]. Due to their two-dimensional nature, the charges excited in a TMDC material can interact strongly via Coulomb interactions. This results in a high exciton binding energy of the order of 0.5 eV [8–10]. Among the TMDCs, molybdenum disulfide (MoS_2) has piqued the interest of scientific community due to its stability, availability, and usage in a wide range of applications [11–13]. In the visible regime, MoS_2 exhibits three significant transitions owing to A, B, and C excitons [7]. In addition to optical properties, MoS_2 exhibits spin and valley-based characteristics [14, 15]. As a result, it is also employed in electronic, spintronic and valleytronic applications [16–18]. Thus understanding the behavior of the carriers excited in MoS_2 under various excited conditions are essential for utilizing it for further applications. Since, A-exciton has its peak at the lowest energy, carriers in an excited ML MoS_2 relax quickly to A-excitonic energy level, resulting a strong

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A-excitonic luminescence [7, 19]. Thus, studies carried out to understand the formation, dissociation and carrier dynamics at A-exciton are crucial for the optoelectronic applications of ML MoS_2.

Relaxation process that happens during the first few picoseconds of an excited carrier is essential in the subsequent evolution of carrier dynamics in materials. Several groups have measured the initial ultrafast response of MoS_2 at excited charge density (N_{ex}) in the range of $\sim 10^{10}$ cm^{-2} to $\sim 10^{14}$ cm^{-2} or equivalently pump fluence ranging from $\sim$0.3 μJ cm^{-2} to $\sim$1.5 mJ cm^{-2} [19–39]. Although, there are overlaps among the regimes in which various processes are anticipated to occur, the relaxation in the initial few picoseconds of time can be categorized into roughly three regimes. For N_{ex} in the range of 10^{10} cm^{-2} to $\sim 10^{11}$ cm^{-2}, the initial rapid decline of the observed optical response is ascribed to the relaxation of excitons via defect states [20, 21] and Auger based recombination assisted by defect states [22–24, 40]. At excitation densities in the range $\sim 10^{11}$ cm^{-2} to $\sim 10^{13}$ cm^{-2}, exciton–exciton annihilation [25–29], defect recombination [19] and carrier cooling through emission of phonons [30–34] are found to play a strong role in the relaxation. At excitation densities higher than $\sim 10^{13}$ cm^{-2} known as the Mott density for MoS_2 [19, 31], the interaction between carriers weakens and hence the exciton–exciton annihilation process becomes insignificant [28, 35]. There are relatively few studies carried out at this high excitation regime [33, 36, 37], but limited to a maximum of 2×10^{14} cm^{-2} [38]. These studies reveal that the relaxation of the carrier occurs via carrier capture by defect states [19–21], defect assisted auger excitation [23, 24], and by hot-photon emission [30, 31].

In this article, we report the work carried out to understand the ultrafast carrier dynamics in a single ML MoS_2 by exciting carriers at A-exciton wavelength at carrier density in the range of $\sim 0.66 \times 10^{14}$ cm^{-2} to 4×10^{14} cm^{-2}. At this experimental condition, it will be shown that an initial fast bandgap renormalization (BGR) is observed which swiftly changed to bleaching within the first few picoseconds. This BGR effect is ascribed to the free carrier creation by exciton dissociation. When a significant percentage of the carriers recombine and the remaining carriers produces excitons, bleaching of this state takes over changing the sign of the observed transient response. These ultrafast carrier dynamic studies are beneficial for applications of ML MoS_2 in devices handling high carrier densities [41–44].

2. Results and discussion

The ML MoS_2 on the sapphire substrate under study was grown by chemical vapor deposition technique [45]. The microscopic optical image of the sample shows several well separated as well as conjoined flakes with well-defined grain boundaries. In figure 1(a), we show the large area absorption spectrum of the sample. The absorption spectrum shows three well distinguishable peaks at 672 nm, 627 nm, and 431 nm, which correspond to the exciton peaks A, B, and C respectively [7, 27, 40, 46]. Using an optical microscope arrangement in the pump–probe setup, the MoS_2 flakes were viewed. Among the various flakes, a specific ML flake was selected for the transient measurements. The optical image of this flake is shown in figure 1(b). To confirm that the selected flake is an ML MoS_2, AFM topography measurement was performed on the same flake and is shown in figure 1(c). Using this topographic image, the thickness of the MoS_2 flake was measured to be $\sim$0.8 nm which is an evidence that the flake on which experiment was performed is an ML [6, 19]. Furthermore, the Raman spectrum of the ML MoS_2 flake, shown in figure 1(d), has peaks at 384 cm^{-1} (in-plane mode E^1_{2g}) and 404.5 cm^{-1} (out-of-plane mode A_{1g}). The difference between these peak wavenumbers is 20.5 cm^{-1}. The peak wavenumbers as well as the difference between them are an indication that the sample studied in the experiments is an ML MoS_2 flake [33, 47].

For performing carrier dynamics measurements, it is essential to confirm that the sample is stable even at much higher excitation conditions than that used in the transient experiments. Pan *et al* investigated the impact of high density laser irradiation on the MoS_2 flakes [37]. They have shown that structural changes in MoS_2 occur when exposed to fluences of the order of 150 mJ cm^{-2}. Exposure to even higher fluences, like 400 mJ cm^{-2} resulted in the development of nanoridges and nanocracks in the MoS_2 flakes [37]. In order to check the stability of the present sample to 672 nm irradiation, we have exposed the ML MoS_2 flakes to different increasing fluences for about 10 min at each fluence. Following each exposure, the AFM image of the same flake was recorded. Figure 2 shows the AFM topographic images of few of these ML MoS_2 flakes after being exposed to different fluences. The topography of the flake remained same even after exposing the sample to fluences up to 50 mJ cm^{-2}. This structural characterization as well as Raman measurement before and after exposure clearly show that the sample is stable for exposure to even higher fluences than that used in the transient measurements (figure S1) (https://stacks.iop.org/CM/34/155401/mmedia). Only when exposed to about 100 mJ cm^{-2}, the AFM image showed a widening of the gap near the grain boundaries of the flakes. Still topography of the ML MoS_2 away from the edges and grain boundaries remained same. Thus even at 100 mJ cm^{-2}, the damage is occurred at the edges and grain boundaries of the flake. In the transient measurements, we have ensured that the maximum fluence used were at least ten times lower than the fluence that could damage the grain boundaries. Moreover, the experiment was also performed at a location well away from the edges and grain boundaries (figure 1(b)).

For performing transient transmission measurements, an OPA pumped by a femtosecond-oscillator-amplifier system (1 kHz, 800 nm, 35 fs) was used. The output of the OPA was tuned near to the A-excitonic peak, 672 nm, of the ML MoS_2. Transient transmission measurements were performed in a standard degenerate pump–probe configuration [48] by pumping and probing at 1.84 eV (672 nm) which is at the A-exciton transition energy. The schematic of the setup is shown in figure 3. The spectral and temporal full width at half maximum (FWHM) of the pulse were measured to be 26 nm and $\sim$60 fs respectively. The FWHM of the pump and

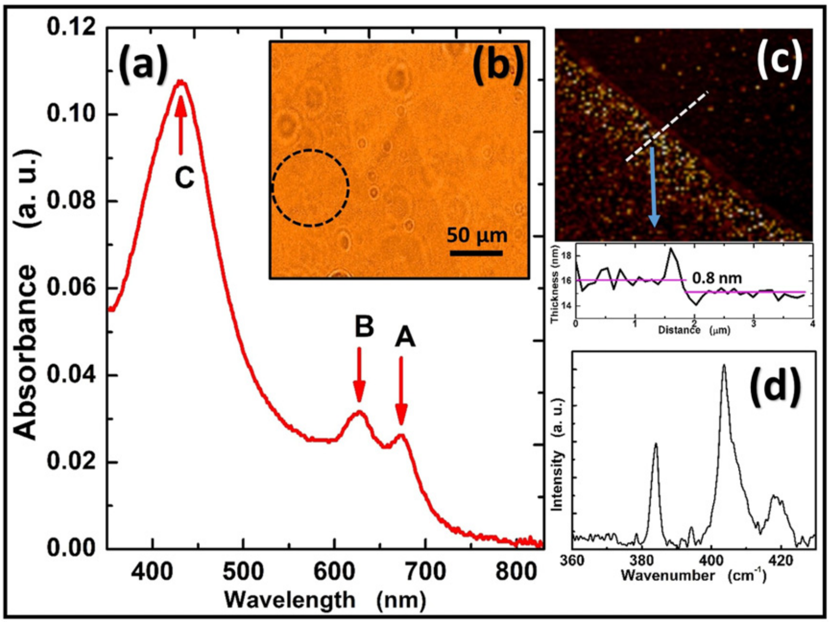

Figure 1. Characterization of the ML MoS_2 sample: (a) absorption spectrum and (b) the optical image of the flake captured in the pump–probe setup using a microscope. The dotted circle shows the location at which the transient experiments were performed. (c) AFM topography image of the MoS_2 flake for the thickness measurement of the MoS_2 flake. The bottom plot shows the height variation along the dotted line. (d) The measured Raman spectrum of the ML MoS_2 flake by exciting at 632.8 nm.

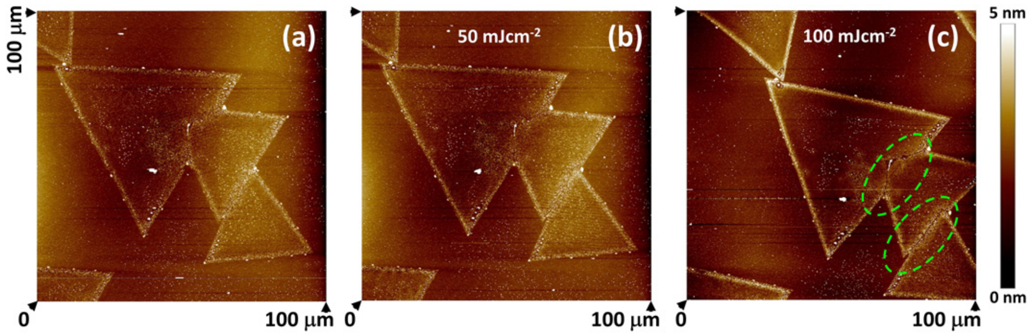

Figure 2. AFM topography image of few monolayer flakes after being exposed to 672 nm femtosecond laser beam at different fluences: (a) before exposure, (b) exposure to 50 mJ cm^{-2} and (c) 100 mJ cm^{-2}. Dotted ellipses shows the regime where structural modifications were observed.

probe beams were measured to be 72 μm and 24 μm respectively. A microscope with a 5× objective, LED light source, and CCD camera was used to photograph the MoS_2 sample placed in a pump–probe setup. The sample was moved while being viewed through this microscope such that the pump and probe spots were focused on to the ML MoS_2 flake at a desired location (figure 1(b)). The dotted circle in the figure shows the location at which the transient experiments were performed. Since the size of the probe beam is much smaller than the flake size, the measurements were performed on a single monolayer flake (figure 1(b)).

The transient transmission response ($\Delta T/T$) of the ML MoS_2 sample near A-exciton wavelength was measured at various pump fluences and is shown in figure 4(a). With the arrival of the pump pulse, the transmission through the sample reduces reaching its maximum change within nearly the duration of the pump beam. As the delay between the pump and the probe pulse increases, the magnitude of change in $\Delta T/T(|\Delta T/T|)$ reduces. At lower pump fluences, i.e. $<\sim$4 mJ cm^{-2} the $|\Delta T/T|$ nearly recovers to zero by about few picoseconds. With the increase in the pump fluence, the maximum change in $|\Delta T/T|$ also increases. At higher fluences ($>$4 mJ cm^{-2}), the $\Delta T/T$ recovers and becomes zero by about 1.2 ps (figure 4(a)). However, with further delay, the sign of $\Delta T/T$ changes. This positive $\Delta T/T$ recovers in tens of picoseconds time scales. Figure 4(c) shows the variation of $\Delta T/T$ in the first one picosecond delay time in log scale. The linear variation of the logarithm of $\Delta T/T$ with delay clearly indicates that the decay could well be approximated to a single exponential in this temporal regime. At 1.2 mJ cm^{-2}, the decay time of this fast initial recovery (τ_f) was estimated to be nearly 0.4 ps. To quantitatively understand the dependence of $\Delta T/T$ on fluence, the variation of the peak change in the magnitude of transmission ($(\Delta T/T)_{pk}$) and τ_f were estimated

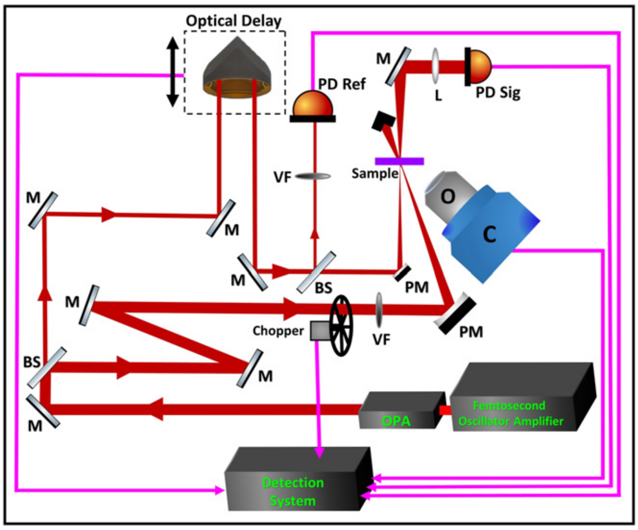

Figure 3. Schematic of the degenerate transient transmission setup used in the measurement. BS: beam splitter; C: CCD camera; L: lens; M: mirror; O: objective; PD: photo-diode; PM: parabolic mirror; and VF: variable filter.

from the time dependence (figure 4(a)). Figure 5(a) shows the dependence of $(\Delta T/T)_{\mathrm{pk}}$ on the pump fluence. We find that the $(\Delta T/T)_{\mathrm{pk}}$ increases linearly with the pump fluence indicating that contribution from higher-order nonlinearities or saturation of signal are negligible. Figure 5(b) shows the fluence dependence of the τ_{f}, which is nearly constant independent of the pump fluence in the present fluence range.

When excited by an ultrafast pulse of photon energy higher than the band gap, carriers are excited from the filled valence band to the conduction band of MoS_2. As discussed in the introduction, several groups have studied the initial recovery of the change in the ultrafast optical response of MoS_2 and demonstrated the physical process which produce the observed changes [19–38, 40]. Various processes like carrier decay to defect states [19–21], defect assisted Auger recombination [22–24, 40], excited-state absorption [35], exciton formation [32–34], exciton–exciton annihilation [25–29], hot-carrier cooling [30, 31], etc are expected to follow after the excitation of carriers. These processes will lead to band filling [49], bleaching [50], BGR [49, 51], valence band broadening [39], exciton peak shift and its broadening [26, 35, 52] etc. These processes modifies the optical response of the material at the probe wavelength. The contribution of each of these process in a given measurement for a given material depends on the wavelength of the pump and probe and also on the pump fluence.

In the present case, we have excited the MoS_2 with a wavelength tuned to the A-exciton, thus the pump pulse will directly create primarily A-excitons [35, 51]. Further, N_{ex} is much above the Mott density hence exciton–exciton annihilation process cannot explain the observed relaxation of the signal [25, 27, 28]. Density of defects in ML MoS_2 are expected to be of the order of 10^{13} cm^{-2}–10^{14} cm^{-2} [53–55]. Thus, a portion of the excitons will relax via defect states causing the initial decay in the signal [19, 20, 49]. However, at the maximum excited carrier densities used in the present experiment (4×10^{14} cm^{-2}) defect states should have been saturated. Further, if the carriers excited to the conduction band are further getting excited by the pump, it should have shown a non-linear variation in the fluence dependence of $(\Delta T/T)_{\mathrm{pk}}$. Wang *et al* and Brasington *et al* observed excitation of carriers from defect states to deep into the conduction band [23, 56]. Since defect states are getting saturated at higher excitation densities, such absorption by defect state should have shown a saturation behavior. This is due to the fact that the number of defects in the sample is expected to be lower than that of the excited carriers at these fluences. The linear relation between fluence and $(\Delta T/T)_{\mathrm{pk}}$ in the experiment, as shown in figure 5(a), clearly shows that there are no saturation or any higher-order nonlinear process. Hence excited state absorption or further absorption from defect states cannot fully explain the observed transient signal. Further, since carriers are directly excited to A-exciton further relaxation of carriers by intraband carrier decay cannot also happen in the present case. However at high densities, excitons are shown to dissociate to form free-carriers [19, 30, 31, 36]. Following such dissociation, the presence of large number of carriers will lead to BGR process [39, 50]. This process reduces the bandgap and will increase the absorption at lower wavelengths [39, 50]. In few layer MoS_2, such dissociation of excitons happens within tens of femtoseconds [36]. Further, in presence of large number of carriers once again Auger and defect assisted Auger process can lift carriers deep into their respective bands thus creating hot carriers [35]. It is well known that defect assisted Auger processes is expected to show a strong N_{ex} dependence [35]. The time dependence of the density of carriers in the MoS_2 following an excitation by a pump pulse is given by [23, 24],

$$N(t) = \frac{N_0}{1 + tN_0\gamma},$$
$$\text{where } \gamma = \frac{\alpha\beta n_{\mathrm{d}}^2}{\alpha n_{\mathrm{d}} + \beta n_{\mathrm{d}}}, \quad (1)$$

$N(t)$ is the remaining excited carrier density in the sample at any time t,N_0 is the initial injected carrier density, α and β are the Auger recombination rates for the electrons and holes respectively and n_{d} is the defect density in the sample. This relation is shown to hold good even when the excited carrier density is more than that of the defects [23, 24]. Using the earlier reported values of α and β and assuming a defect density of 5×10^{13} cm^{-2} in ML MoS_2 [53–55], we find that the defect assisted Auger based recombination should also happen in few tens of femtoseconds time scales. Since the pulse width of the pump is nearly 60 fs, the exciton dissociation as well as the subsequent excitation of carriers deeper into the conduction band via the Auger process should also occur within the duration of the pump pulse. Thus, we propose that by the end of the pump pulse the BGR, some carriers are relaxed, and some amount of hot carriers are also present in the sample. A schematic of the proposed process is shown in figure 6(a).

Over time, the hot-carriers present in the ML at the end of the pump pulse will relax to the edge of their respective

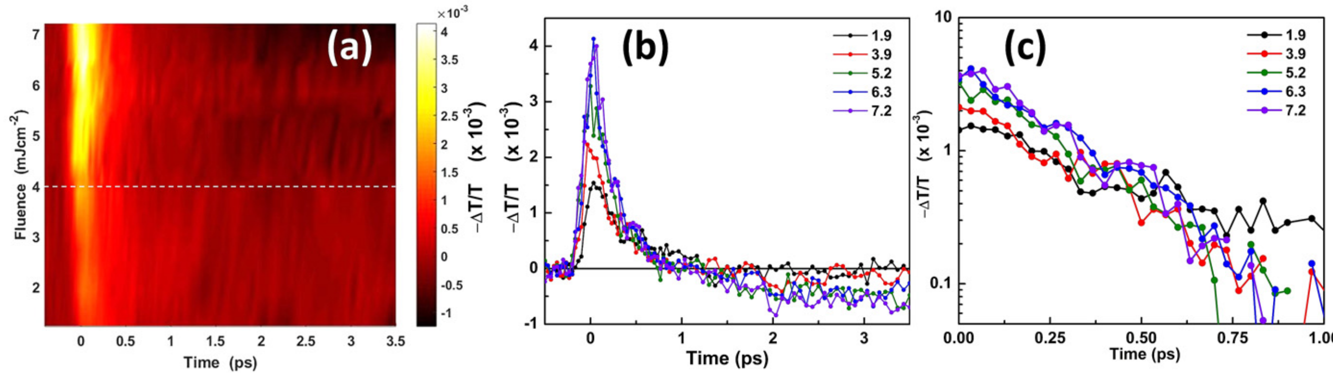

Figure 4. Transient transmission signal measured on the ML MoS_2 flake at its A-exciton peak at various pump fluences (note the negative sign of the $\Delta T/T$ in the plot axis): (a) 2D pseudocolor image of the transient transmission data expressed in $\Delta T/T$ as a function of both pump fluence and delay time, (b) some representative plots of transient transmission signals at few fluences. The legend shows the pump fluences in mJ cm^{-2} and (c) the same transient signal in log scale zoomed at the first one picosecond delay regime.

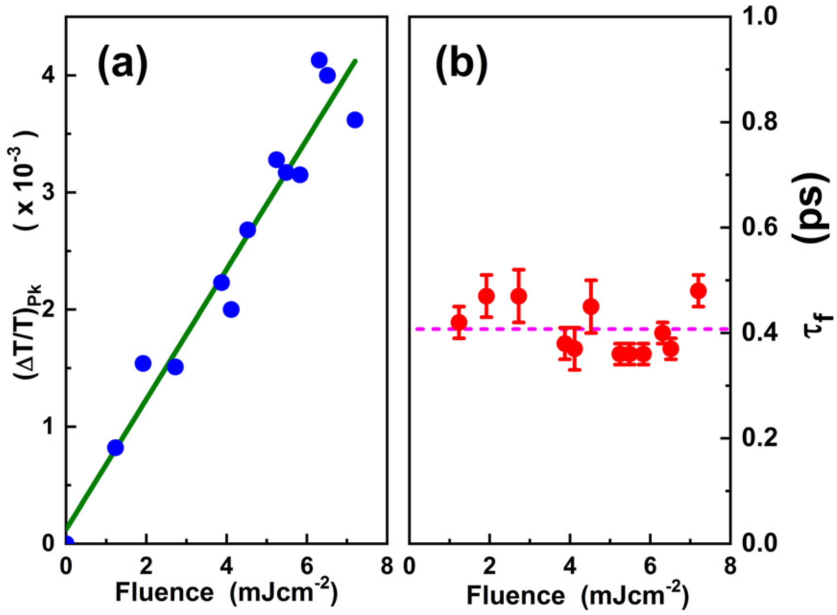

Figure 5. (a) Variation of the peak change in magnitude of $\Delta T/T$ with pump fluence extracted from the 2D plot shown in figure 4(a). The linear fit to the experimental points is also shown as a solid line. (b) Shows the fluence dependence of the initial fast decay time. The average decay time is shown as a dotted line.

band extrema by emission of phonons. Although such relaxation time depends on how deep the carriers are excited into their respective bands such hot-carrier cooling time is reported to be in the order of few hundreds of femtoseconds [36]. Furthermore, the relaxation of carriers to the defect state is also expected to happen at similar time scales [19, 20]. Due to Auger scattering as well as the relaxation of carriers to the defect states, the overall N_{ex} present in the ML would eventually reduce and hence, the probability for the formation of exciton will increase over time. Thus, it is expected that over time A-excitons will form leading to the bleaching of this state. Such bleaching will give a positive contribution to $\Delta T/T$ at the probe wavelength reducing the effect of BGR. Thus, as the delay between the pump and probe increases, the initial observed negative $\Delta T/T$ will start recovering over time. The relaxation time for hot-carriers, formation time for exciton and relaxation to defect states are all expected show fluence independent relaxations similar to that observed here(figure 5(b)) [23, 24, 30, 31]. Figure 6(b) shows a schematic of the process of the initial relaxation of carriers proposed above.

Having discussed the initial negative $\Delta T/T$ signal and its recovery, let us now look at the origin of the positive $\Delta T/T$ signal (figure 4). At lower pump fluences (<4 mJ cm^{-2}) the observed initial negative change in transmission eventually recover back to the unperturbed condition. However, at fluences more than 4 mJ cm^{-2}, the $\Delta T/T$ signal changes and becomes positive. If the excitation density is high enough, the overall heat generated in the sample can lead to a bottleneck effect increasing the life time of carriers in the A-exciton state [35, 49]. Further, the defect states will also saturate at higher excitation densities leaving some A-excitons to live longer [49]. Thus, we attribute the observed positive $\Delta T/T$ in the

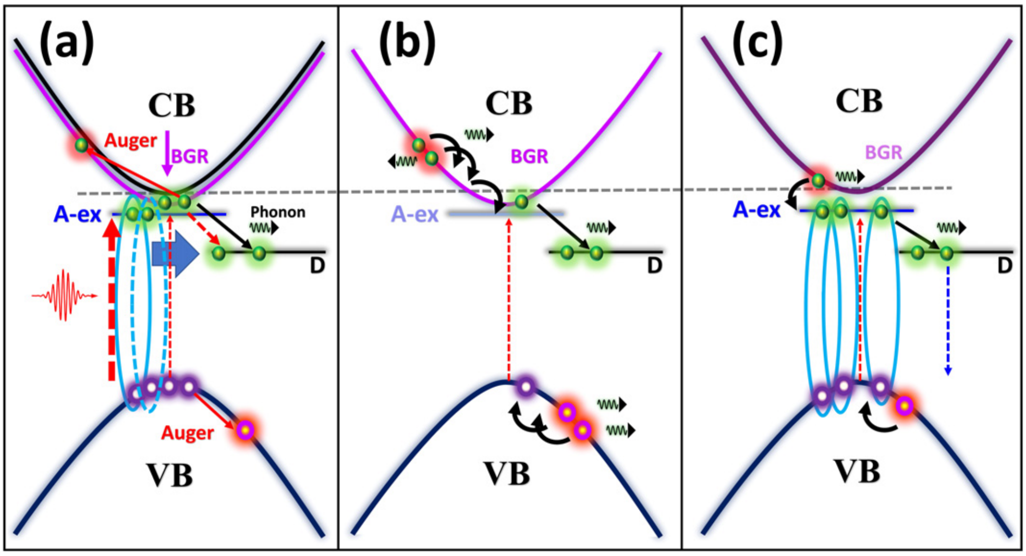

Figure 6. Schematic of the proposed processes: (a) exciton dissociation leads to BGR and creation of hot-carrier by Auger and defect assisted Auger scattering. (b) Cooling of hot-carriers leading to formation of exciton and relaxation of carriers to defect states which causes the relaxation of the measured $\Delta T/T$. (c) Filling of A-excitonic state leading to bleaching.

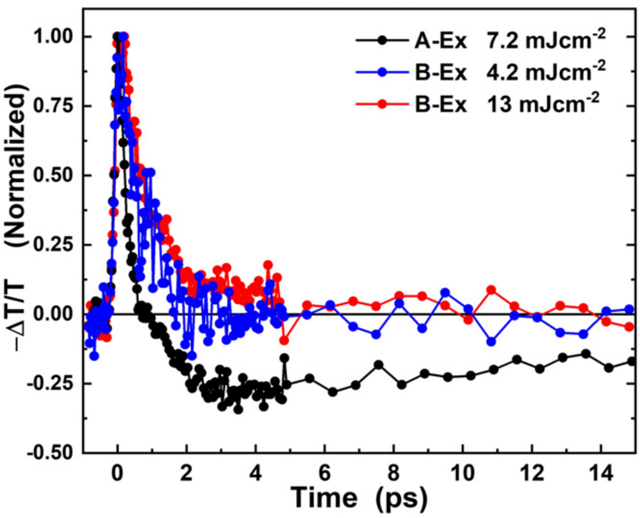

Figure 7. Degenerate transient transmission signal measured on the same ML MoS_2 flake at its A-exciton (A-Ex) and B-exciton (B-Ex) peaks. Only the measurement at A-excitonic wavelength shows bleaching effect.

experiment to a bleaching effect that originate from the bottleneck effect (schematic: figure 6(c)). Such bottleneck effect will not occur when probed at a higher photon energies. This is due to the fact that carriers at higher energy states will eventually decay to A-exciton [19, 52].

A transient measurement at a different wavelength well above the A-exciton like that close to B-exciton is needed to conform the proposed process. Figure 7 shows the normalized transient transmission signal measured on the same MoS_2 flake but at B-exciton wavelength. For comparison, the transient transmission signal measured at A-exciton is also shown along with that of B-excitonic response. The measurement at B-exciton wavelength also showed a negative $\Delta T/T$ which is what expected when BGR is in action [39, 50]. Even at double the fluence than that used for A-exciton measurement we find that the $\Delta T/T$ measured at B-exciton did not show any positive signal. This indicates that at these fluences only BGR dominates and bleaching effects does not show up as that expected in the proposed model.

3. Conclusions

We have provided a comprehensive analysis of the ultrafast optical response of ML MoS_2 at higher excitation densities due to its rising demand in optoelectronic applications. The ultrafast optical response of ML MoS_2 is investigated at the least studied excitation densities in the range of $\sim$0.66 $\times 10^{14}$ cm^{-2} to 4 $\times 10^{14}$ cm^{-2} using carrier dynamics measurements. Damage threshold measurements revealed that the topology of ML MoS_2 gets modified at the grain boundaries at fluences of the order of 100 mJ cm^{-2}. In ultrafast transient studies, it was discovered that excited carriers create an initial BGR effect, which quickly switches to bleaching. This BGR effect is caused by the generation of free carriers by exciton dissociation, which is subsequently followed by defect-assisted Auger scattering. During this transition period, majority of the charge carriers recombine, while the remaining form excitons, allowing bleaching to take over at later times. Our studies and findings will be helpful for the application of ML MoS_2 as a component in lasers [41], compact OPAs [42], and high power detectors [43, 44] where high excitation densities are involved.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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